

Experiment -1

Name: Gurveer Singh Cheema **SAP ID: 590032925**

Programme: B.Tech. CSE **Year:** 1st Year **Batch:** 15

Professor: Dr. Christalin Nelson

1. Setting up a platform to work on LINUX

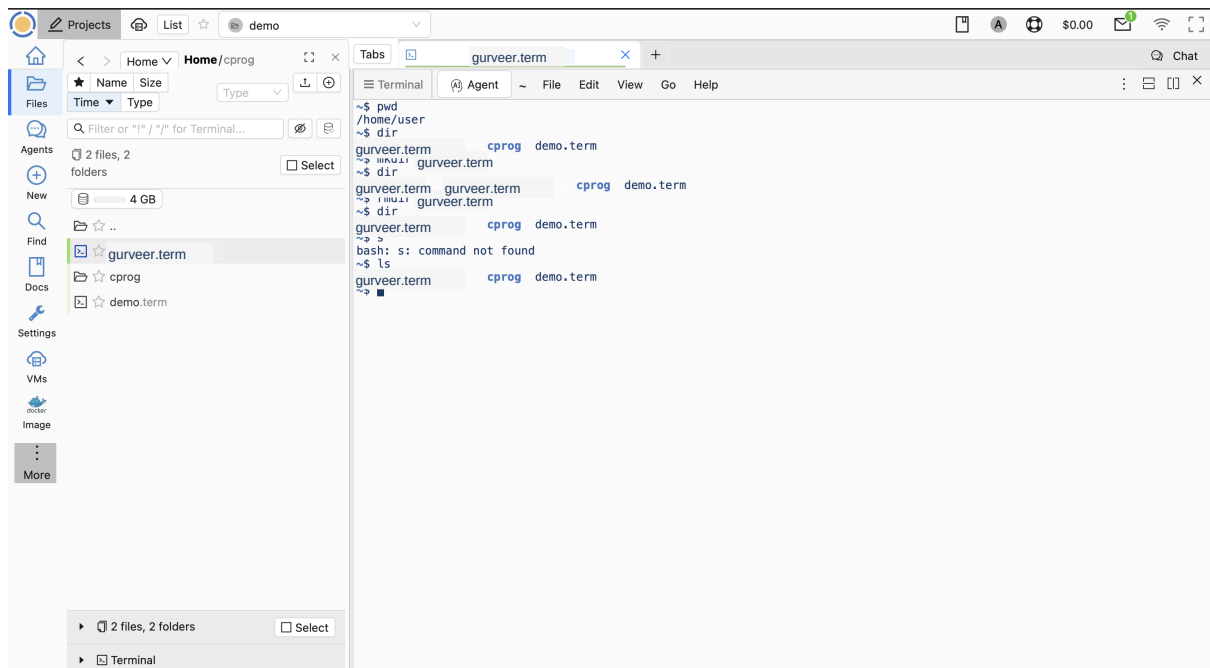
There are two ways to work on LINUX

1. Offline
2. Online

To use LINUX online we can use websites like

1. COCALC
2. Cuddy
3. Distro sea

In our case we used COCALC to work on LINUX



To use LINUX offline there are multiple ways

1. Dual boot
2. Single boot
3. Bootable Pendrive
4. Virtual Machine

We will be using virtual machine to work on LINUX(VM)

Virtual Machine Manager (Hypervisor)

Definition: A virtual machine manager, or hypervisor, is software that creates and manages virtual machines by allocating the host computer's processor, memory, storage, and other hardware resources to guest operating systems.

Type-1 (Baremetal)

Definition: A Type-1 (bare-metal) hypervisor runs directly on the physical hardware rather than on a conventional host operating system. It manages the hardware and the guest operating systems.

Examples: VMware ESXi, Microsoft Hyper-V, Xen, and KVM.

Type-2 (Hosted)

Definition: A Type-2 (hosted) hypervisor runs as an application on top of a host operating system such as Windows, Linux, or macOS. The host operating system handles access to the physical hardware.

Examples: Oracle VirtualBox, VMware Workstation, VMware Fusion, and Parallels Desktop.

For this experiment, Oracle VirtualBox is used as the Type-2 hypervisor.

Installation of VMM & VM

Step-1: Download Oracle VirtualBox from the official downloads page:

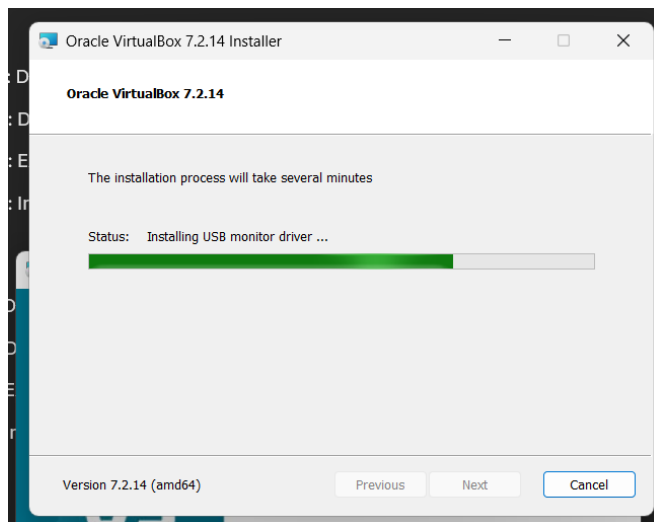
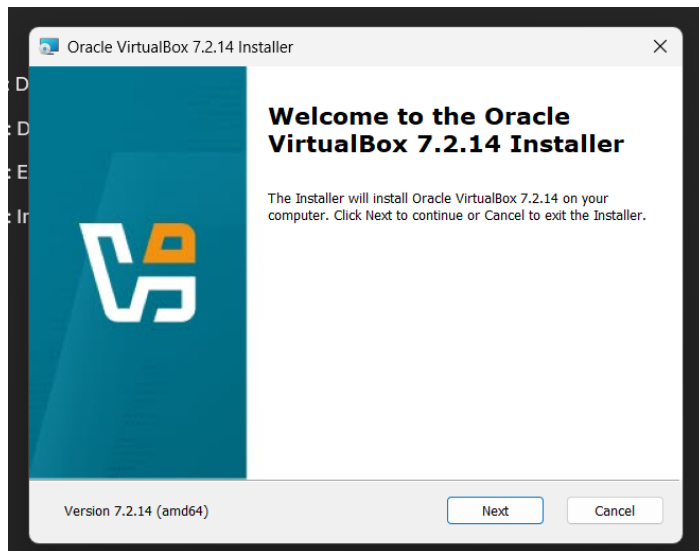
<https://www.virtualbox.org/wiki/Downloads>

Step-2: Download the pre-built Kali Linux image for VirtualBox from:

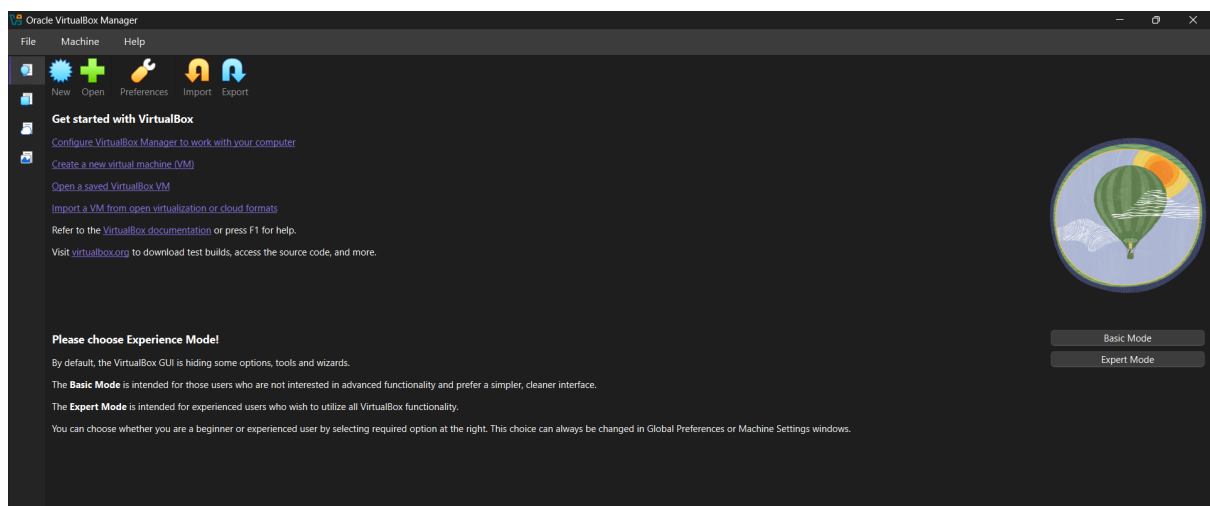
<https://www.kali.org/get-kali/#kali-virtual-machines>

Step-3: Extract the downloaded Kali Linux archive to a suitable folder. The extracted folder contains the VirtualBox virtual-machine files.

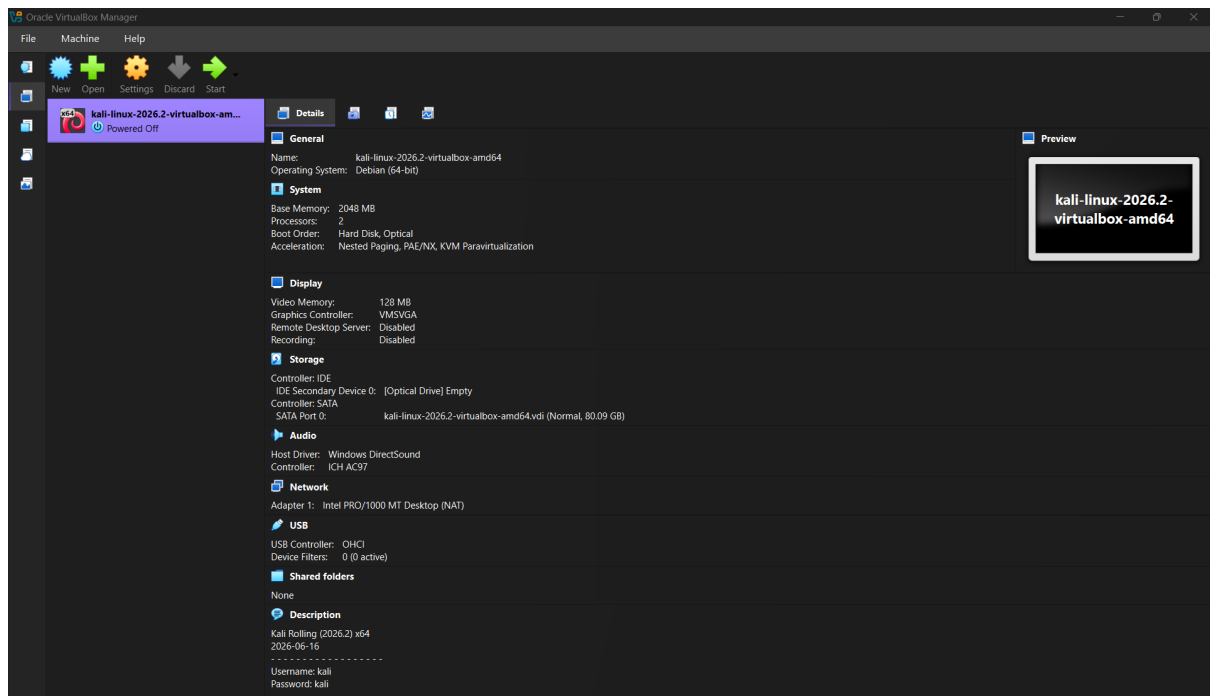
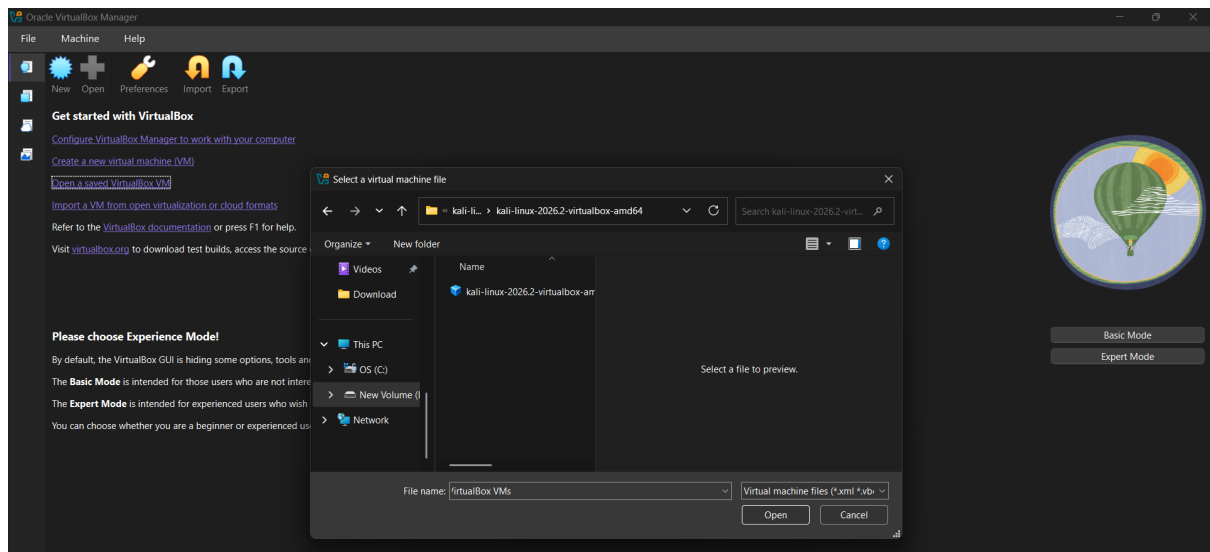
Step-4: Run the Oracle VirtualBox installer and follow the on-screen instructions to finish the installation.



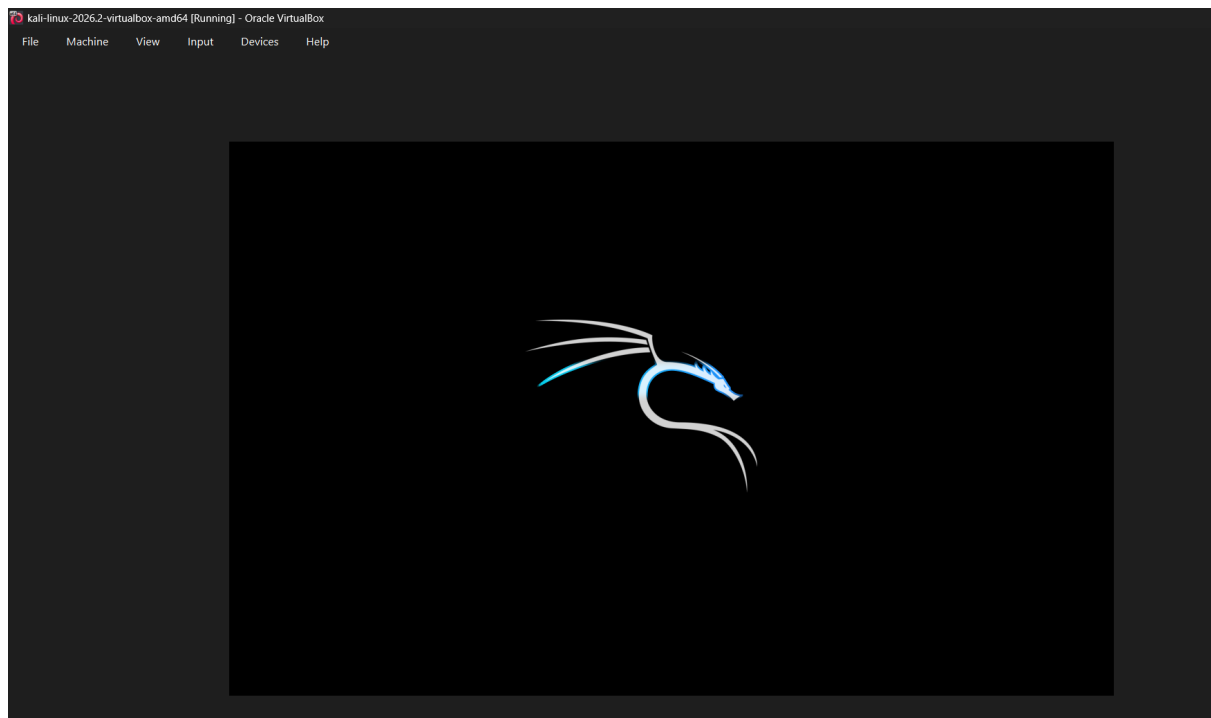
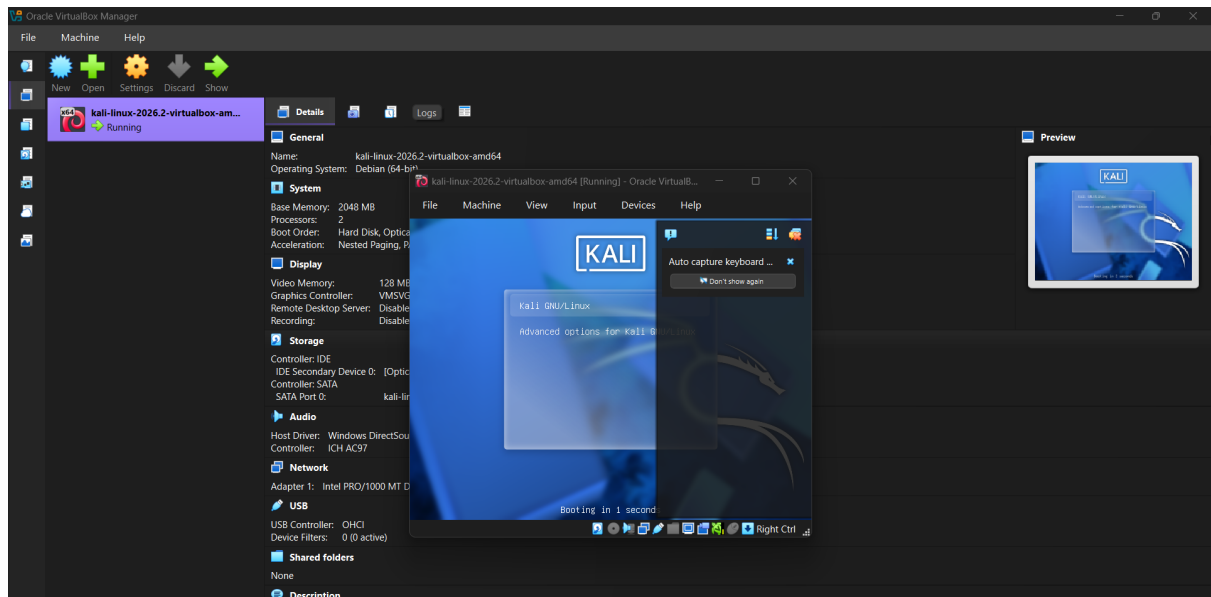
Step-5: Open Oracle VirtualBox Manager.

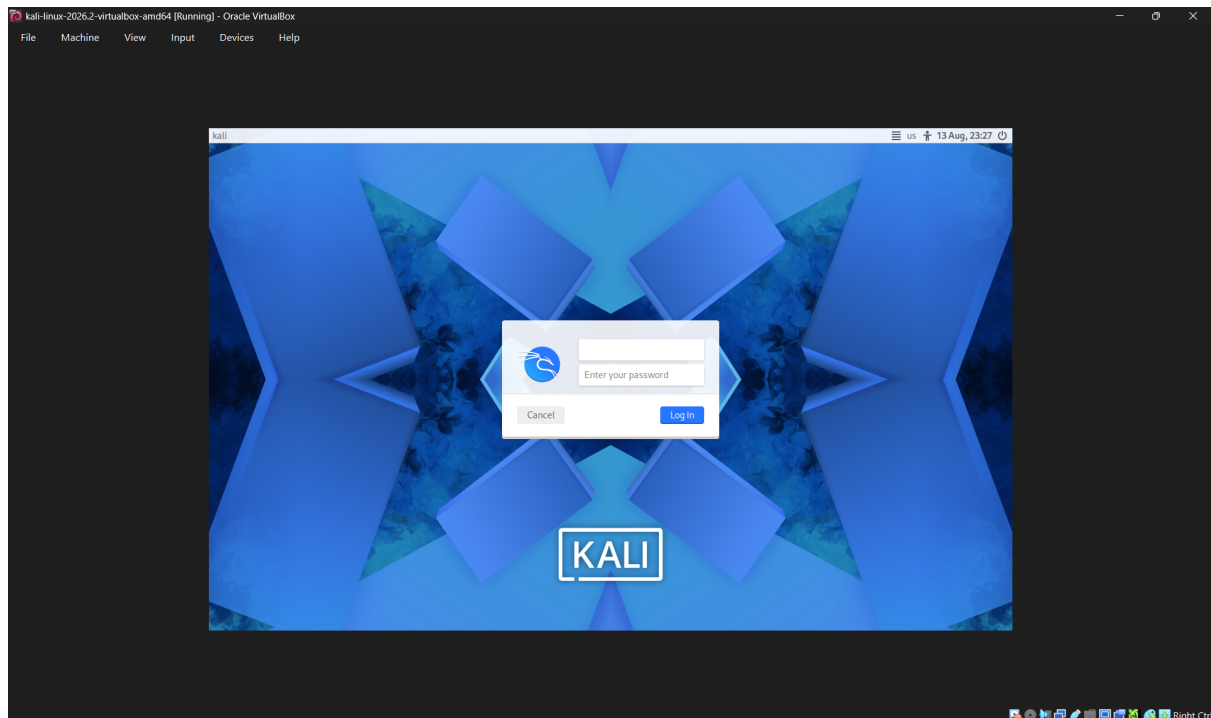


Step-6: Select Open or Add, browse to the extracted Kali Linux .vbox file, and open it to register the saved virtual machine in VirtualBox.



Step-7: Select the Kali Linux virtual machine and click Start. At the login screen, use the default username kali and password kali, and then change the password if instructed.





Result: The Type-2 virtual machine manager (Oracle VirtualBox) and the Kali Linux virtual machine were installed, registered, and started successfully

For MAC:

On MAC you can do it offline via the terminal app which is already installed in the system